Extended Abstract: Diagnosing Small-Business Life Cycles from Weekly Tran

Research Question

Do small businesses actually follow the conventional start-up-growth-maturity-decline life cycle, or do weekly transaction records reveal more heterogeneous paths? We examine whether high-frequency sales data can define economically meaningful life-cycle states for small businesses and whether early signals such as sales trend, drawdown risk, volatility, and new-customer acquisition help distinguish recovery from persistent decline.

Methodology

This study uses weekly transaction records from Korean Credit Data (KCD) for 59,089 establishments in Seoul's food-service sector. We combine two complementary empirical designs. First, for post-entry dynamics, we study recently opened establishments by aligning stores by weeks since opening and recovering recurring sales-path patterns from weekly time series. We then summarize movement before and after major turning points to assign interpretable trajectory labels such as DDZ and UUX. Second, for the broader stock of surviving businesses, we use an observed-window design covering 50,635 establishments with at least 52 observed weeks and classify current Growth, Stable, and Decline states across the full sample.

We then estimate multinomial logit models to examine which variables separate these states across business-age buckets. The main explanatory variables include sales trend, maximum drawdown (MDD), volatility, new-customer ratio, delivery share, and industry and local-market controls. Finally, we test whether long-run outcomes can be predicted from only the first 30 weeks of observed operation by comparing baseline models with specifications that incorporate trajectory-based features.

Expected Results

Three findings stand out. First, post-entry trajectories are highly heterogeneous. In the 108-week entrant sample, the dominant pattern is persistent decline followed by exit-stage behavior (DDZ), indicating that the canonical hump-shaped life cycle is a poor summary of most establishments. Second, the full-sample observed-window analysis yields a different but complementary picture: among 50,635 establishments, 40.0 percent are classified as Growth, 38.2 percent as Stable, and 21.7 percent as Decline. This distribution should be interpreted as the pattern of currently observed windows among surviving stores rather than as a literal description of the initial life cycle of all businesses.

Third, the variables that separate outcomes are not constant over age. Sales trend is the most consistent discriminator across age groups, maximum drawdown is closely associated with decline, and new-customer ratio becomes increasingly informative for Growth in older establishments. In the observed-window age-bucket models, the Growth share rises and the Decline share falls in older groups, while early-stage groups show a more even split between Growth, Stable, and Decline. In complementary forecasting exercises, adding trajectory-based features to baseline store characteristics materially improves predictive performance over level-only specifications, showing that early path information contains meaningful signal about later outcomes.

Significance

This study contributes to the economics of small-business dynamics by replacing survey-based or purely age-based life-cycle definitions with transaction-based diagnoses grounded in observed behavior. It also shows that two empirical lenses are necessary: entry-cohort analysis for early post-opening dynamics and observed-window analysis for the current state of the full surviving store population. For policy, the implication is that support should rely less on broad linear stages and more on observed trajectory type, trend deterioration, drawdown risk, and customer-acquisition dynamics when identifying businesses that are likely to recover versus those facing persistent decline.